

Fama–MacBeth Cross-Sectional Analysis of Variance Risk Pricing in Chinese A-Share Markets

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1 Introduction

This paper tests whether variance risk premium (VRP) exposures are priced in the cross-section of A-share stock returns using a two-pass Fama–MacBeth framework, with 16 VRP variants tested across daily, weekly, and monthly frequencies. Following the BTZ volatility regime classification, we find that cross-sectional VRP pricing is highly state-dependent, concentrated in high-volatility periods and most pronounced at the weekly frequency.

2 Methodology

2.1 Factor Construction

Market Factor (MKT). The market return is the value-weighted average of all individual stock returns each period:

$$\text{MKT}_t = \frac{\sum_i \text{MktCap}_{i,t-1} \cdot r_{i,t}}{\sum_i \text{MktCap}_{i,t-1}}. \quad (1)$$

Daily returns including dividend reinvestment (**Dretwd**) and total market capitalization (**Dsmvt11**) are sourced from CSMAR. For weekly and monthly frequencies, individual stock returns are compounded from daily observations.

Size (SIZE). Size is the natural logarithm of each stock’s total market capitalization:

$$\text{SIZE}_{i,t} = \ln(\text{MktCap}_{i,t}). \quad (2)$$

Book-to-Market Ratio (B/M). Following the Fama–French timing convention, book value from the December annual report of year $y - 1$ is matched to trading months from July of year y through June of year $y + 1$. The ratio is then:

$$\text{B/M}_{i,t} = \frac{\text{BookValue}_{i,y-1}}{\text{MktCap}_{i,t}}, \quad (3)$$

winsorized at the 1st and 99th percentiles each period.

2.2 Volatility Regime Classification

Market states are defined using the BTZ framework. Let μ_t and σ_t denote the rolling mean and standard deviation of the China VIX over a backward-looking window. The three regimes are:

$$\text{Low (Complacency): } \text{VIX}_t < \mu_t - \sigma_t, \quad (4)$$

$$\text{Medium (Normal): } \mu_t - \sigma_t \leq \text{VIX}_t \leq \mu_t + \sigma_t, \quad (5)$$

$$\text{High (Panic): } \text{VIX}_t > \mu_t + \sigma_t. \quad (6)$$

Rolling windows of 252 days, 52 weeks, and 36 months are used for daily, weekly, and monthly frequencies respectively.

2.3 Fama–MacBeth Regression

First pass. For each stock i , factor exposures are estimated via rolling time-series regression:

$$r_{i,\tau} = \alpha_i + \beta_i^{\text{VRP}} \cdot \text{VRP}_\tau + \beta_i^{\text{MKT}} \cdot \text{MKT}_\tau + \varepsilon_{i,\tau}. \quad (7)$$

Second pass. At each period t , a cross-sectional regression is estimated across all stocks:

$$r_{i,t} = \lambda_0^t + \lambda_{\text{VRP}}^t \hat{\beta}_i^{\text{VRP}} + \lambda_{\text{MKT}}^t \hat{\beta}_i^{\text{MKT}} + \lambda_{\text{SIZE}}^t \text{SIZE}_{i,t} + \lambda_{\text{B/M}}^t \text{B/M}_{i,t} + \eta_{i,t}. \quad (8)$$

Inference. The Fama–MacBeth risk premium estimate and its t -statistic are:

$$\hat{\lambda} = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}^t, \quad t = \frac{\hat{\lambda}}{\text{std}(\hat{\lambda}^t)/\sqrt{T}}. \quad (9)$$

3 Experiments

The empirical analysis is structured around three dimensions of variation.

VRP variants. We test 16 transformations of the base VRP measure. These fall into four broad categories: (i) basic transformations (`vrp_squared`, `vrp_cubic`, `vrp_log_ratio`); (ii) standardizations and normalizations (`vrp_normalized`, `vrp_zscore`, `vrp_abs`); (iii) signed decompositions and interaction terms (`vrp_positive`, `vrp_negative`, `vrp_x_rv`, `vrp_x_vix`, `iv_rv_spread`, `iv_rv_spread_std`); and (iv) momentum-based measures (`vrp_change_1d`, `vrp_mom_5d`, `vrp_acceleration`). Each variant is used separately as the VRP factor in the first-pass time-series regression.

Return frequencies. Each variant is tested at three return frequencies: daily, weekly, and monthly. Beta estimation windows are set to 252 trading days, 52 weeks, and 36 months respectively, with a minimum of half the window length required. Weekly and monthly returns are compounded from daily data; VRP variants are aggregated as within-period means.

Volatility regimes. For each frequency and variant, the full-sample Fama–MacBeth regression is re-estimated separately within each of the three BTZ volatility regimes (low, medium, high), yielding $16 \times 3 \times 4 = 192$ sets of coefficient estimates in total. This design allows us to assess whether the cross-sectional pricing of VRP risk is state-dependent, as suggested by the insurance-selling interpretation of variance strategies.

4 Empirical Results

Tables 1–3 report the estimated $\hat{\lambda}_{\text{VRP}}$ for all 16 VRP variants across the four sample partitions (full sample and three regimes) at each frequency. Rows are ordered by the absolute t -statistic in the full weekly sample. Bold entries indicate statistical significance at the 10% level or better.

Table 1: Fama–MacBeth $\hat{\lambda}_{\text{VRP}}$ by Regime: Daily Frequency

VRP Variant	All	Low	Medium	High
vrp_x_rv	−2.329	3.143	−3.133	−7.363
vrp_mom_5d	−0.589***	−0.626**	−0.531**	−0.709
vrp_zscore	−0.105***	−0.096*	−0.095**	−0.144
vrp_acceleration	0.134	−0.054	0.006	0.770*
vrp_change_1d	0.020	−0.147	−0.071	0.516
vrp_log_ratio	−0.012	−0.014	−0.007	−0.025
iv_rv_spread_std	−0.006	−0.007	−0.003	−0.012
vrp_negative	0.010	−0.004	0.007	0.038
vrp_positive	−0.010	0.004	−0.007	−0.038
iv_rv_spread	−0.029	−0.041	−0.018	−0.049
vrp_normalized	−0.010	−0.015	−0.004	−0.021
vrp	−0.293	−0.465	−0.193	−0.358
vrp_cubic	−255.056	35.046	−563.176*	270.109
vrp_x_vix	−5.491	−10.396	−3.828	−3.789
vrp_abs	−0.134	−0.399*	−0.042	−0.049
vrp_squared	−2.890	−8.075	−5.166	10.942

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ Table 2: Fama–MacBeth $\hat{\lambda}_{\text{VRP}}$ by Regime: Weekly Frequency

VRP Variant	All	Low	Medium	High
vrp_x_rv	−3.773***	−0.955	−3.073**	−9.928***
vrp_mom_5d	−0.526***	−0.546**	−0.192	−1.462**
vrp_zscore	−0.085**	−0.055	−0.041	−0.257**
vrp_acceleration	−0.140**	−0.195**	−0.227***	0.190*
vrp_change_1d	−0.134**	−0.215**	−0.094	−0.127
vrp_log_ratio	−0.017*	−0.006	−0.002	−0.077**
iv_rv_spread_std	−0.009*	−0.003	−0.001	−0.042**
vrp_negative	0.025*	−0.011	0.018	0.096**
vrp_positive	−0.025*	0.011	−0.018	−0.096**
iv_rv_spread	−0.044*	−0.020	−0.005	−0.195**
vrp_normalized	−0.019*	−0.008	0.001	−0.092**
vrp	−0.467*	−0.289	−0.025	−2.003**
vrp_cubic	−459.137*	−213.838	−128.787	−1772.922
vrp_x_vix	−10.338*	−9.256	0.419	−42.981**
vrp_abs	−0.236	−0.477*	0.113	−0.890
vrp_squared	−8.150	−10.431	0.240	−29.027

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Fama–MacBeth $\hat{\lambda}_{\text{VRP}}$ by Regime: Monthly Frequency

VRP Variant	All	Low	Medium	High
vrp_x_rv	−0.210	−0.042	1.072	− 18.302**
vrp_mom_5d	−0.184	−0.052	−0.206	−0.677
vrp_zscore	−0.018	0.116*	−0.038	−0.550
vrp_acceleration	0.044*	0.052**	0.037	0.088
vrp_change_1d	0.022	0.036	0.010	0.095
vrp_log_ratio	0.018	−0.005	0.006	0.317
iv_rv_spread_std	0.009	−0.003	0.003	0.163
vrp_negative	−0.028	−0.010	−0.010	−0.370
vrp_positive	0.028	0.010	0.010	0.370
iv_rv_spread	0.056	−0.011	0.028	0.835
vrp_normalized	0.019	−0.006	0.007	0.344
vrp	0.707	−0.080	0.442	8.969
vrp_cubic	477.832	176.419	255.940	5244.863**
vrp_x_vix	17.447	0.024	12.700	185.277
vrp_abs	0.434	0.190	0.425	2.020
vrp_squared	11.655	6.961	7.970	88.959*

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5 Analysis

Weekly frequency dominates. The weekly frequency produces the most consistent and significant results across variants and regimes (Table 2). Thirteen out of 16 variants are significant at the 10% level or better in at least one regime, compared to five variants at daily and three at monthly frequency. This suggests that a one-week horizon is the natural frequency at which VRP information is incorporated into stock prices.

Strong regime dependence. Across all frequencies, significance is concentrated in the high-volatility (Panic) regime. At the weekly frequency, 12 variants are significant in the high regime, compared to 5 in the full sample and 3 in the low regime. This pattern is consistent with the BTZ hypothesis that VRP strategies function as insurance-selling mechanisms that are only activated during periods of elevated market stress.

Sign of the risk premium. The estimated $\hat{\lambda}_{\text{VRP}}$ is negative across nearly all significant specifications. Under the standard VRP definition (IV − RV), this implies that stocks with higher VRP beta earn lower returns in the cross-section.